THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solution turns both red and blue litmus paper to red. The solution is: 1

(a) basic **(b)** neutral **(c)** acidic **(d)** salty

2. In $HCl + NaOH \longrightarrow NaCl + H_2O$, the base is: 1

(a) HCl **(b)** H_2O **(c)** $NaCl$ **(d)** $NaOH$

3. Why does vinegar taste sour? 1

(a) it contains acetic acid **(b)** it contains salt

(c) it contains sugar **(d)** it contains sodium

4. A solution that turns red litmus blue and turmeric red is: 1

(a) basic **(b)** sugary **(c)** acidic **(d)** neutral

5. In $H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$, the salt formed is: 1

(a) sodium sulphate **(b)** sodium carbonate **(c)** sodium nitrate **(d)** sodium chloride

6. Why does baking soda taste bitter? 1

(a) it is a salt **(b)** it is a base **(c)** it is acid **(d)** it is sugar

7. A solution does not change the colour of litmus or turmeric. It is: 1

(a) basic **(b)** salty only **(c)** neutral **(d)** acidic

8. In $HNO_3 + KOH \longrightarrow KNO_3 + H_2O$, the salt formed is: 1

(a) sodium nitrate **(b)** potassium sulphate

(c) potassium chloride **(d)** potassium nitrate

9. Why does curd become sour on keeping long? 1

(a) water evaporates **(b)** sugar is added

(c) more lactic acid forms **(d)** it becomes basic

10. A student finds that China rose turns green with a solution. The solution is likely: 1

(a) lemon juice **(b)** vinegar **(c)** soap solution **(d)** curd

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A solution turns red litmus blue. What does it contain? Give an example. 2

12. Write a neutralisation reaction of sulphuric acid with sodium hydroxide. 2

13. Differentiate between a salt and an acid. 2

14. Why is vinegar, and not baking soda, applied on a sting that is alkaline in nature (such as that of a wasp)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A solution X turns red cabbage extract green and turmeric red. Another solution Y turns it red and turmeric stays yellow. Identify X and Y. 3

16. Why are some stomach pills chewed and others dissolved? 3

17. Why should chemicals in the lab be labelled? 3

OR

Describe how to prepare a litmus-like indicator from beetroot or red cabbage and test with vinegar and soap. 3

18. Explain why a solution of copper sulphate cannot be stored in an iron vessel but acidic pickles also react with metals. 3

OR

Give three examples of acids and bases used in cooking. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. You are given five unlabelled liquids: vinegar, lime water, sugar solution, lemon juice and soap solution. Design a test to identify each. 5

OR

A student mixes 10 mL of vinegar with baking soda. Describe the observation and explain the reaction in words. 5

20. Describe the full activity of neutralisation of hydrochloric acid with sodium hydroxide using litmus, with equation. 5

OR

Explain the effect of indicators on a series of substances and conclude on a rule for acids and bases. 5

END OF QUESTION PAPER — Total: 40 Marks

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